# JEE Apex FTS – JEE Main Full Syllabus Test-05

## KEY SHEET

## MATHEMATICS

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | **3** | 2 | **4** | 3 | **1** | 4 | **4** | 5 | **2** |
| 6 | **2** | 7 | **1** | 8 | **3** | 9 | **3** | 10 | **2** |
| 11 | **2** | 12 | **1** | 13 | **1** | 14 | **3** | 15 | **3** |
| 16 | **3** | 17 | **1** | 18 | **4** | 19 | **3** | 20 | **2** |
| 21 | **125** | 22 | **27** | 23 | **1** | 24 | **80** | 25 | **3** |

## PHYSICS

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 26 | **2** | 27 | **1** | 28 | **3** | 29 | **3** | 30 | **1** |
| 31 | **4** | 32 | **1** | 33 | **1** | 34 | **3** | 35 | **1** |
| 36 | **3** | 37 | **4** | 38 | **4** | 39 | **2** | 40 | **4** |
| 41 | **2** | 42 | **4** | 43 | **4** | 44 | **1** | 45 | **4** |
| 49 | **2** | 47 | **2** | 48 | **2** | 49 | **8** | 50 | **2** |

## CHEMISTRY

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 51 | **1** | 52 | **2** | 53 | **3** | 54 | **4** | 55 | **1** |
| 56 | **1** | 57 | **3** | 58 | **3** | 59 | **2** | 60 | **2** |
| 61 | **1** | 62 | **1** | 63 | **1** | 64 | **2** | 65 | **1** |
| 66 | **3** | 67 | **3** | 68 | **4** | 69 | **1** | 70 | **1** |
| 71 | **3** | 72 | **2** | 73 | **4** | 74 | **36** | 75 | **14** |

# SOLUTION
# MATHEMATICS

**01.** A, B, C , D are coplanar $\Rightarrow \left[\overline{AB}, \overline{AC}, \overline{AD}\right] = 0$

$$\begin{vmatrix} -4 & -3 & 3-2\lambda \\ -7 & \lambda-5 & 4-2\lambda \\ -6 & 0 & 6-2\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 5\lambda + 6 = 0 \Rightarrow \lambda = 2,3$$

$$\sum_{\lambda \in 5} (\lambda+3)^2 = 5^2 + 6^2 = 61$$

**02.** $3x^2 - 4y^2 = 36$ & $\dfrac{dy}{dx} = \dfrac{3x}{4y}$

$$\left(\frac{dy}{dx}\right)_{(x_0,y_0)} = \frac{3x_0}{4y_0} = \frac{-3}{2} \Rightarrow x_0 = -2y_0$$

$$3x_0^2 - 4y_0^2 = 36 \Rightarrow y_0 = \pm\frac{3}{\sqrt{2}} \text{ then } x_0 = \mp\frac{6}{\sqrt{2}}$$

$(x_0, y_0) = \left(\dfrac{-6}{\sqrt{2}}, \dfrac{3}{\sqrt{2}}\right), \left(\dfrac{6}{\sqrt{2}}, \dfrac{-3}{\sqrt{2}}\right)$ Calculating perpendicular distance from both coordinates

then $\left(\dfrac{6}{\sqrt{2}}, \dfrac{-3}{\sqrt{2}}\right)$ is nearest to line

**03.** $y^2 = 4\sqrt{5}x$

Equation of any tangent is $y = mx + \dfrac{\sqrt{5}}{m}$

This is tangent to $x^2 + y^2 = \dfrac{5}{2}$

$$\Rightarrow \frac{\left|\frac{\sqrt{5}}{m}\right|}{\sqrt{1+m^2}} = \frac{\sqrt{5}}{\sqrt{2}} \Rightarrow m^4 + m^2 - 2 = 0 \Rightarrow m = \pm 1$$

Equation of common tangent is $y = x + \sqrt{5}$

Also $m = \pm 1$ Satisfies $m^4 - 3m^2 + 2 = 0$

**04.** $\alpha, \beta, \gamma$ are roots of $x^3 - x^2 - 1 = 0$. then $\alpha + \beta + \gamma = 1$, $\alpha\beta + \beta\gamma + \gamma\alpha = 0$

$\alpha^2 + \beta^2 + \gamma^2 = 1$ also $\beta^3 = \beta^2 + 1$, $\gamma^4 = \gamma^3 + \gamma = \gamma^2 + \gamma + 1$

Now $\alpha^2 + \beta^3 + \gamma^4 = \alpha^2 + \beta^2 + 1 + \gamma^2 + \gamma + 1 = 3 + \gamma$

Let $3 + \gamma = x \Rightarrow \gamma = x - 3$ then $f(\gamma) = 0$

Replace $x$ by $x - 3$ in given equation $\Rightarrow x^3 - 10x^2 + 33x - 37 = 0$

**05.** $(22)^{2022} + (2022)^{22}$

Divided by $3 \Rightarrow (21+1)^{2022} + (3 \times 674 + 0)^2 = 3k + 1$ remainder $\alpha = 1$

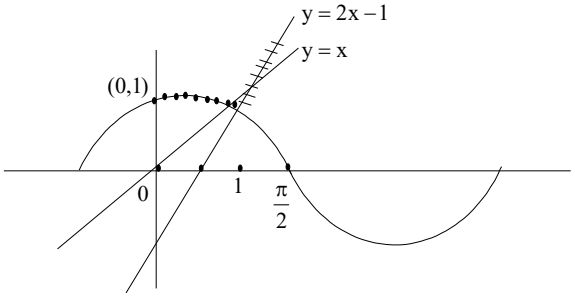
Divided by $7 \Rightarrow (21+1)^{2022} + (7\times 289-1)^{22} = (7\lambda_1+1)+(7\lambda_2+1) = 7\lambda+2$ Remainder $\beta = 2$

**06.** **(A)** $\lim\limits_{h\to 0} \dfrac{f'(1)(-3\sinh+4\cosh)}{f'(2)\left(3e^h-5\sec h+\tanh\right)} = \dfrac{6\times 4}{2\times 3} = 4$

**(B)** $\lim\limits_{h\to 0} \dfrac{2(2a+h)-f(2a)}{h} = 4 \Rightarrow f'(2a) = 4$

Now $\lim\limits_{h\to 0} \dfrac{f(-2a+h)-f(-2a)}{2h} = \dfrac{f'(-2a)}{2} = \dfrac{f'(2a)}{2} = 4$

(C) $\dfrac{G'(0)}{F'(0)} = \dfrac{e^{f(0)}.f^1(0)}{f'(1).e^0} = \dfrac{e^3.3}{e^3} = 3$




No of non differentiabilityof f(x) = 2
(since $f(x) = f(4a-x)$ in $[2a,4a]$ and f is even equation)

**07.** $\dfrac{dy}{dx} - \dfrac{y}{x} = y^3(1+\log x) \Rightarrow \dfrac{1}{y^3}\dfrac{dy}{dx} - \dfrac{1}{xy^2} = 1+\log x$

Let $\dfrac{-1}{y^2} = t \Rightarrow \dfrac{2}{y^3}\dfrac{dy}{dx} = \dfrac{dt}{dx}\ \dfrac{dt}{dx} + \dfrac{2t}{x} = 2(1+\log x)$ then I.F $= e^{\int \frac{2}{x}dx} = x^2$

$t.x^2 = \int 2x^2(1+\log x)dx \Rightarrow \dfrac{-x^2}{y^2} = \dfrac{2}{3}\left[(1+\log x)x^3 - \dfrac{x^3}{3}\right] + C \qquad y(1) = 3 \Rightarrow C = \dfrac{-1}{3}$

**08.** $n(A) = 48,\ n(B) = 25\ n(C) = 18$

$n(A\cup B\cup C) = 60,\ n(A\cap B\cap C) = 5$

$n(A\cup B\cup C) = \sum n(A) - \sum n(A\cap B) + n(A\cup B\cup C)$

$\Rightarrow \sum n(A\cap B) = 48+25+18+5-60 = 36$

No of men who received exactly 2 medals $= \sum n(A\cap B) - 3n(A\cap B\cap C) = 36-15 = 21$

**09.** $t_r = \sqrt{1+\dfrac{1}{r^2}+\dfrac{1}{(r+1)^2}} = \sqrt{\dfrac{r^2(r+1)^2+(r+1)^2+r^2}{r^2(r+1)^2}}$

$= \sqrt{\dfrac{r^4+2r^3+3r^2+2r+1}{r^2(r+1)^2}} = \dfrac{r^2+r+1}{r(r-1)} = 1+\dfrac{1}{r(r+1)} = 1+\dfrac{1}{r}-\dfrac{1}{r+1}$

$S = 2000 - \dfrac{1}{2000} \Rightarrow \left|2000(s-2000)\right| = 1$

**10.** $n_1 = n_2 = 5 \qquad \bar{x}_1 = 5,\ \bar{x}_2 = 8 \qquad \sigma_1^2 = 12,\quad \sigma_2^2 = 20$

$n_3 = 10, \quad \bar{X}_1 = \frac{\sum(x_i - 3)}{5} = \frac{\sum x_i}{5} - \frac{3\times 5}{5} = 5 - 3 = 2$

$\bar{X}_2 = \frac{\sum(x_j + 2)}{5} = \frac{\sum x_j}{5} + \frac{2\times 5}{5} = 8 + 2 = 10$

Combined mean $= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{10 + 50}{10} = 6$

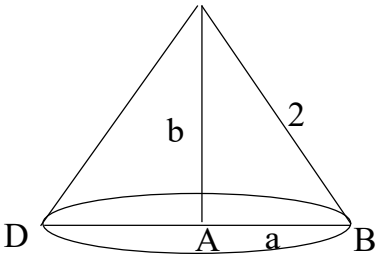
But variance is not change by change of origin

**11.** Let $\frac{xy}{x+y} = n \left(n = 2^3 3^4 5^6\right) \Rightarrow xy = nx + ny \qquad \Rightarrow (x-n)(y-n) = n^2 = 2^6 3^8 5^{12}$

No of positive integral solutions = no of positive divisors of $n^2 = 2^6 3^8 5^{12}$
$= (6+1)(8+1)(12+1) = 819$

**12.** From $\Delta ABC$, $a^2 + b^2 = 4$.....(1)

Volume $V = \frac{1}{3}\pi a^2 b = \frac{1}{3}\pi b\left(4 - b^2\right)$




$V' = 0 \Rightarrow b = 2/\sqrt{3}$

$V'' < 0$ at $b = 2/\sqrt{3} \quad \therefore V$ is maximum

Maximum volume $V = \frac{16\pi}{9\sqrt{3}}$

**13.** $\frac{\sum x_i}{6} = 2$ and $\frac{\sum x_i^2}{6} - 2^2 = 23 \quad \Rightarrow \alpha + \beta = 10$ and $\alpha^2 + \beta^2 = 52$

Solving we get $\alpha = 4,\ \beta = 6 \qquad$ M.D $= \frac{\sum|x_i - \bar{x}|}{6} = \frac{13}{3}$

**14.** By kings rule $f(1-x) = -f(x) \Rightarrow I = 0$

**15.** $\frac{x+6}{3} = \frac{y}{2} = \frac{z+1}{1} = r$ and $\frac{x-7}{4} = \frac{y-9}{3} - \frac{z-4}{2} = s$

At point of intersection on $x = 3r - 6 = 4s + 7$.....(1)

$y = 2r = 3s + 9 \rightarrow (2)$

$z = r - 1 = 2s + 4 \rightarrow (3)$

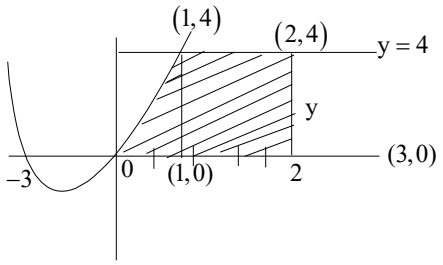
By solving we get r = 3 P.O.I $= (3, 6, 2)$

**16.** P (drawing white ball) $= \frac{1}{2} =$ P(getting black ball)

P (drawing white ball 4th time in 7th drawn) $= {}^6C_3\left(\frac{1}{2}\right)^3 \cdot \left(\frac{1}{2}\right)^3 \cdot \frac{1}{2} = \frac{5}{32}$

**17.** $y \le x^2 + 3x,\ 0 \le x \le 3,\ 0, \le y \le 4$

$y = 4 \Rightarrow x^2 + 3x = 4 \Rightarrow x = 1, -4$




$$R.A = \int_0^1 \left(x^2 + 3x\right)dx + \int_1^3 4dx$$

**18.** $\Delta = \begin{vmatrix} \lambda & 2 & 2 \\ 2\lambda & 3 & 5 \\ 4 & \lambda & 6 \end{vmatrix} = -(\lambda + 8)(\lambda - 2) \quad \Delta_1 = \begin{vmatrix} 5 & 2 & 2 \\ 8 & 3 & 5 \\ 10 & \lambda & 6 \end{vmatrix} = -9\lambda + 34$

For $\lambda = 2 \Rightarrow \Delta = 0$ and $\Delta_1 \neq 0$ For $\lambda = 2$ no solution

**19.** $\cos\theta = \frac{1}{\sqrt{5}}$ then $\sin 2\theta = \frac{4}{5} \Rightarrow \alpha = 1$ $\sin A = \frac{4}{5}, \cos B = \frac{3}{5}$ then $\beta = \cos(A + B) = \frac{-7}{25}$

Req equation $25x^2 - 18x - 7 = 0$

**20.** $A = \{1,2,3\}, B = \{1,2,3,4,5,6,7,8\}$

$f(i) \le f(j)$ for all $i < j$

$f(1) = 1,\ f(2) = 1,\ f(3) = \{1,2,3,.....8\} \rightarrow 8$way

$f(2) = 2,\ f(3) = \{2,3.....8\} \rightarrow 7$ way

And…….so on

$f(2) = 8\ , f(3) = \{8\} \rightarrow 1$way

Now $f(1) = 1$ then no of functions formed $= 8 + 7 + 6 + ..... + 1 = \sum 8$

$f(1) = 2\ \ f(2) = 2\ , f(3) = \{2,3.....8\} \rightarrow 7$ways

$f(2) = 3\ \ f(3) = \{3,4,.....\} \rightarrow 6$ways

And ……so on

$f(2) = 8\quad f(3) = 8 \rightarrow 1$way

$f(1) = 2$ then no of functions formed $= 7 + 6 + ..... + 1 = \sum 7$

Similarly $f(1) = 3$ then no of functions formed $= \sum 6$

$f(1) = 4,5,6,7,8$ formed functions $= \sum 5, \sum 4, \sum 3, \sum 2,$ & $\sum 1$ respectively

Total functions formed $= \sum 8 + \sum 7 + ..... + 1 = 120$

**21.** $2(A + B) = A^T + B^T + 3I.....(i)$

Transpose on both sides

$2\left(A^T + B^T\right) = A + B + 3I$

$2\left(A^T + B^T\right) = \left(\frac{A^T + B^T + 3I}{2}\right) + 3I.....$from (i)

$\Rightarrow A^T + B^T = 3I.....(ii)$

Given $AA^T = 4I \Rightarrow A\left(3I - B^T\right) = 4I$

$\Rightarrow 3A - AB^T = 4I$ (transpose on both sides we get)

$\Rightarrow 3A^T - BA^T = 4I \Rightarrow BA^T = 3A^T - 4I$.....(iii)

Also $AA^T = 4I \Rightarrow A^T = 4A^{-1} \Rightarrow 12A^{-1} = 3A^T$.....(iv)

Now $\left|12A^{-1} - BA^T + I\right| = \left|3A^T - \left(3A^T - 4I\right) + I\right| = |5I| = 5^3.|I| = 125$

**22.** Since $Z = 1 + i$ and $Z_1 = \dfrac{1 + i\bar{z}}{\bar{z}(1 - z) + \dfrac{1}{z}} = \dfrac{1 + i(1 - i)}{(1 - i)(-i) + \dfrac{1}{1 + i}} = -1 + i$ $\text{Arg}(z_1) = \dfrac{3\pi}{4}$

**23.** $A(3,10)$

Image of $B(7,2)$ w.r.t the line $2x + y - 6 = 0$ is $B'(-1,-2)$

Equation of $AB^1$ is $3x - y + 1 = 0$ $\quad a = 3, b = -1$

**24.** $e_1e_2 = 1 \Rightarrow \sqrt{\dfrac{25 - b^2}{25}}.\sqrt{\dfrac{16 + b^2}{16}} = 1 \quad \left(25 - b^2\right)\left(16 + b^2\right) = 25 \times 16$

$400 + 9b^2 - b^4 = 400 \Rightarrow b^2 = 9$

$\Rightarrow e_1 = \dfrac{4}{5}$, $e_2 = \dfrac{5}{4}, \alpha = 2ae_1 = 8, \beta = 2ae_2 = 10$

**25.** $I = \int \dfrac{\sin x}{\cos^3 x(\tan^3 x + 1)}dx = \int \dfrac{\tan x \sec^2 x}{(\tan x + 1)\left(\tan^2 x - \tan x + 1\right)}dx$

Let $\tan x = t \Rightarrow \sec^2 x \, dx = dt$

$I = \int \dfrac{tdt}{(t + 1)(t^2 - t + 1)}$

Let $\dfrac{t}{(t + 1)\left(t^2 - t + 1\right)} = \dfrac{A}{t + 1} + \dfrac{B(2t - 1)}{t^2 - t + 1} + \dfrac{C}{t^2 - t + 1}$

$t = A(t^2 - t + 1) + B(2t - 1)(t + 1) + C(t + 1)$

Compare like terms, we get $A = \dfrac{-1}{3}$, $B = \dfrac{1}{6}$, $C = \dfrac{1}{2}$

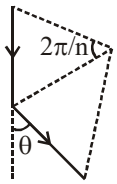
$I = \dfrac{-1}{3}\int \dfrac{dt}{t + 1} + \dfrac{1}{6}\int \dfrac{2t - 1}{t^2 - t + 1}dt + \dfrac{1}{2}\int \dfrac{dt}{t^2 - t + 1}$

$= \dfrac{-1}{3}\log_e|1 + \tan x| + \dfrac{1}{6}\log_e\left|\tan^2 x - \tan x + 1\right| + \dfrac{1}{2}.\dfrac{2}{\sqrt{3}}\tan^{-1}\left|\left(\dfrac{2\tan - 1}{\sqrt{3}}\right)\right| + C$

$\alpha = \dfrac{-1}{3}$, $\beta = \dfrac{1}{6}$, $\gamma = \dfrac{1}{\sqrt{3}}$

# PHYSICS

**26.** Sides are shown




$$\theta = \pi - 2\left(\frac{\pi - \frac{2\pi}{n}}{2}\right) = \frac{2\pi}{n}$$

Here n = 12 $\Rightarrow \theta = \frac{2\pi}{12} = \frac{\pi}{6}$

Velocity of approach = V – V cos $\frac{\pi}{6} = V - \frac{\sqrt{3}}{2}V$

Time of approach $= \frac{a}{V\left(1 - \frac{\sqrt{3}}{2}\right)}$

Distance travelled $= \left(\frac{2a}{2-\sqrt{3}}\right)$.

**27.** $T - 50 = 5(4) \quad T = 70N$

$T + N = 200 \quad N = 200 - 70 \quad = 130\ N$

**28.** $avg\ velocity = \frac{change\ in\ velocity}{time\ taken}\ \frac{\sqrt{2}\,v}{3/4\,t}\ where\ v = \frac{2\pi R}{t}$

**29.** Velocity of the mass $m$ when the string is just taut $v_o = \sqrt{2g(2h)}$

From impulse momentum considerations,

$$(M+m)V = mv_o \qquad \therefore \qquad V = \frac{2m\sqrt{gh}}{M+m}$$

**30.** Conceptual

**31.** Conceptual

**32.** Fdt = $mv_0$ $\quad (Fdt)h = \frac{2}{5}mv_0R \quad$ Solving $h = \frac{2R}{5}$

**33.** When the oil is poured, the fraction of ice in the water decreases, i.e., volume of ice melted into water is greater than volume of water displaced by ice. So water level rises. Overall volume of ice will decrease as it melts. So the upper level of oil falls

**34.** Velocity of efflux. $v = \sqrt{2gh}$

Velocity of top layer $= \frac{AV}{Av} = \frac{A}{A_0}\sqrt{2gh}$

$\therefore$ acceleration top layer $= \frac{d}{dx}\left[\frac{A}{Av}\sqrt{2gh}\right] \qquad \frac{A}{Av}\sqrt{2g} \times \frac{1}{2\sqrt{h}} \times \frac{dh}{dt}$

$= \frac{A}{A_0}\sqrt{2g} \times \frac{1}{2\sqrt{h}} \times \frac{A}{A_0}\sqrt{2gh} \quad \left(\frac{A}{A_0}\right)^2 g$

Correct option is (c

**35.** A-q, B-p, C-p, D-q

Hint: $g = \frac{Gm}{R^2} = \frac{G\left(\frac{4}{3}\pi R^3\right)\rho}{R^2}$ Or $g \propto \rho R$

At centre, $V = -\frac{Gm}{R} = -\frac{G\left(\frac{4}{3}\pi R^3 \rho\right)}{R}$ or $V \propto \rho R^2$

**36.** CONCEPTUAL

**37.** Let temperature of interface is θ

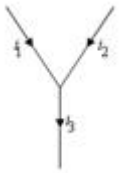
$$\frac{\theta - 0}{\frac{1}{4\pi K}\left[\frac{1}{R} - \frac{1}{2R}\right]} = \frac{70 - \theta}{\frac{1}{4\pi(2K)}\left[\frac{1}{2R} - \frac{1}{3R}\right]}$$

$$\frac{\theta}{\frac{1}{R}} = \frac{70-\theta}{\frac{1}{6R}} \quad \theta = 6(70-\theta)$$

$7\theta = 6 \times 70$ $\theta = 60^{\circ}C$

**38.** Electric flux $\oint_s \vec{E}.d\vec{s} = \frac{q_{in}}{\varepsilon_o}$

**39.**



$$\frac{v_1 - v_0}{R_1} + \frac{v_2 - v_0}{R_2} = \frac{v_0 - v_3}{R_3}$$

**40.** Conceptual

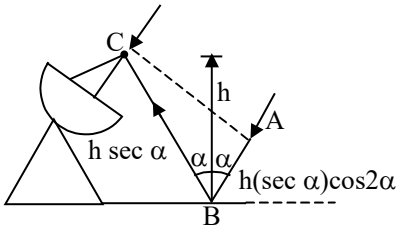
**41.** $\int_a^{3a} \frac{\mu_0 IL}{2\pi}\left(\frac{dx}{x} + \frac{dx}{4a-x}\right) = \frac{\mu_0 IL}{2\pi}\left[\ln x - \ln(4a-x)\right]_a^{3a} = \frac{\mu_0 IL \ln 3}{\pi}$

**42.** $P = \int \frac{\varepsilon^2}{R} = \int_0^a \frac{\left(\pi r^2 \omega B_0 \cos\omega t\right)^2 b dr}{2\pi r \rho} = \frac{1}{8\rho}\pi b a^4 \omega^2 B_0^2 \cos^2 \omega t$ $P_w = \frac{1}{16\rho}\pi b a^4 \omega^2 B_0^2$.

**43.** Intensity $= \frac{\text{no of photons incident}}{\text{second} \times \text{Area}}$, current $\alpha \frac{\text{no. of photons incident}}{\text{seconds}}$

**44.** CONCEPTUAL

**45.** Total path difference = AB + BC + λ/2 = λ for maxima




$h \sec\alpha \cos 2\alpha + h \sec\alpha = \lambda/2$

$h \sec\alpha\left(2\cos^2\alpha\right) = \lambda/2$

$$h = \frac{\lambda}{4\cos\alpha}$$

**46.** Let v is the volume of balls

$$V(\rho_1 + \rho_2)g = (6\pi\eta rv)2 - - - (1)$$

$$V\rho_1 g - T = 6\pi\eta rv - - - (2)$$

Breaking stress $= \frac{T}{A}$ $= \frac{\left(\frac{1}{4}\right)10(9-5)10^3}{0.5\times10^{-4}} = 2\times10^8$

**47.** Here, the stretching force, i.e tension in the plank is not the same through out the length of the plank. Let F be the force at a distance x from one end. Then $\frac{F}{x} = \frac{F_0}{l}$ $(or)$ $F = F_0\frac{x}{l}$

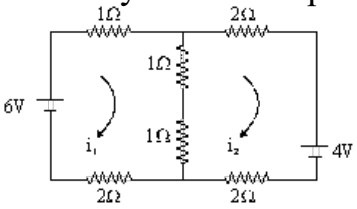
$$Y = \frac{stress}{strain} \qquad \frac{f}{s} = Y \times strain$$

$$\frac{f_0 x}{ls} = Y \times strain \qquad Y \times \frac{increase\ \ in\ length\ dx}{dx}$$

Increase in $dx = \frac{f_0}{lsY}x\,dx$ Total increase $\Delta l = \frac{f_0}{lsY}\int_0^l x dx = \frac{1}{2}\frac{f_0}{lsY}l^2$ Find $\frac{\Delta l}{l}$

**48.** $\left(\frac{\mu_1}{\mu_0} - 1\right)t_1 = \left(\frac{\mu_2}{\mu_0} - 1\right)t_2$

**49.** In steady state the equivalent circuit is




$$5i_1 - 2i_2 = 6$$

$$6i_2 - 2i_1 = -4$$

$$13i_2 = 14 \quad i_1 = \frac{14}{13}\text{amp},\ i_2 = -\frac{4}{13}\text{amp}$$

Potential difference across BC

$$V_A = 6 - \frac{14}{13} = \frac{64}{13}V$$

$$V_D = \frac{64}{13} - 2\left(-\frac{4}{13}\right) = \frac{72}{13}V$$

$$V_B = \frac{64}{13} - 1\times\left\{\frac{14}{13} + \frac{4}{13}\right\} = \frac{46}{13}V$$

$$V_{DB} = \frac{72-46}{13} = \frac{26}{13} = 2\text{volt}$$

Charge on capacitor = $CV_{BD} = 4\times10^{-6}\times2 = 8\mu C$

**50.** the induced field = $E = BV$

And electric field surface of sphere $= \frac{kQ}{R^2}$

$R = 1.732 \times 10^5$

# CHEMISTRY

**51.** $K_2SO_4Al_2(SO_4)_3 24H_2O$

Total number of +ve charges (or) –ve charges = 8

**52.** 8 Chiral carbons (Glucose + Fructose)

Maltose (Glucose +Glucose)

**53.** $\%N = \dfrac{1.4 \times N \times V}{\text{weight of O.C}} = \dfrac{1.4(50 \times 0.5 \times 2 - 60 \times 0.5 \times 1)}{0.5} = 56$

**54.** Both are isostructural so isomorphous electronic configuration $Xe\,4f^{14}5d^{10}6s^2 6p^1$

Reason: Inert pair effect

**55.** $Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

$\dfrac{V_1M_1}{n_1} = \dfrac{V_2M_2}{n_2} = \dfrac{V_1M_1}{1} = \dfrac{20 \times 0.1}{1} = V_1M_1 = 2$

M moles $Na_2CO_3$ in 250ml $= \dfrac{250}{25} \times = 20$

Weight of $Na_2CO_3 = \dfrac{20 \times 106}{1000} = 2.12$

$\%Na_2CO_3 = \dfrac{2.12}{5} \times 100 = 42.4$

$\%Na_2CO_3 = 100 - 42.4 = 57.6$

**56.** $B \rightarrow$ black solid

$B \rightarrow$ crystalline and amorphous

$C \rightarrow \underset{B}{2.0} \quad \underset{Al}{1.5} \quad \underset{Ga}{1.6} \quad \underset{In}{1.7} \quad \underset{T\ell}{1.8}$

$D \rightarrow$ B is Icosahedron

$E \rightarrow$ Ga liquid metal

**57.**

$$C-C-C \equiv CH \xrightarrow{Na} C-C-C \equiv C.Na$$

$\downarrow$

Trans isomer

With Lindlards $\rightarrow$ cis $Na / NH_3 \rightarrow$ trans

**58.** Unununnium i $\rightarrow$ iii $\rightarrow$ Rontageium

**59.** $2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 10CO_3 + 8H_2O$

$2M\ KMnO_4 = 5M\ C_2O_4^{--}$

$0.1 KMnO_4 = \quad 0.25$

$6KMnO_4 + 10FeC_2O_4 + 24H_2SO_4 \rightarrow 3K_2SO_4 + 6MnSO_4 + 20CO_2 + 5Fe_2(SO_4)_3 + 24H_2O$

$6M KMnO_4 = 10 mol\ FeC_2O_4$

$0.1 \qquad\qquad 0.166$

$2KMnO_4 + 10FeSO_4 + 8H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O$

$2M\ KMnO_4 \quad = 10FeSO_4$

$0.1 \quad = 0.5\ FeSO_4$

**60.** $NH_4 \rightarrow 238.5K$

$PH_3 \rightarrow 185.5K$

$AsH_3 \rightarrow 210K$

$SbH_3 \rightarrow 254.6K$

$BiH_3 \rightarrow 290K$

**61.** **R:** $CH_3CH_2OH$ is $1^\circ$ not give Lucas test due to primay alcohol

**A: correct**

**62.** $m = \frac{1.2}{60} \times \frac{1000 \times 1.06}{2000} = 0.0106$

$\Delta T_f = K_f \times m = 1.86 \times 0.0106 = 0.01976$

$i = \frac{\text{observed}}{\text{expected}} = \frac{-0.0205}{-0.0197} = 1.041$

$AcOH \rightleftharpoons AcO^- + H^+$

$1 \qquad 0 \qquad 0$

$1-\alpha \qquad \alpha \qquad \alpha$

$n = (1+\alpha)$

$i = \frac{n(1+\alpha)}{n} \Rightarrow 1.041 = 1+\alpha \Rightarrow \alpha = 0.041$

$\text{concentration} = C\alpha \Rightarrow 0.0106 \times 0.041 = 4.24 \times 10^{-4}\ m/L$

**63.** $r = k[A]^\alpha [B_2]^\beta \rightarrow (1)$

$1.6 \times 10^{-4} = [0.5]^\alpha [0.5]^\beta \rightarrow (2)$

$3.2 \times 10^{-4} = [0.5]^\alpha [0.5]^\beta \rightarrow (3)$

$3.2 \times 10^{-4} = [1.0]^\alpha [1.0]^\beta \rightarrow (4)$

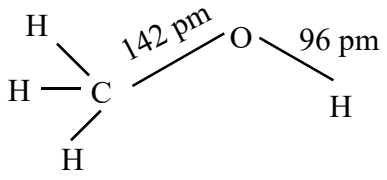
$\frac{2}{3} \Rightarrow \frac{1}{2} = \frac{1}{2^\beta} \quad \text{so, } \beta = 1$

**64.** $Cr_2O_7^{--} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

$6F \Rightarrow 6 \times 96500 = 579000$

$\text{centimole} = \frac{579000}{100} = 5790 \text{ coulombs}$

**65.**




**66.** $B_{12}$ It is stored in the body

**67.** Amino acids (Biomolecules)

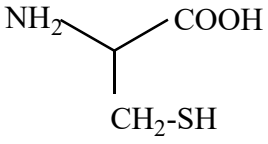
Tryptophan – W
Lysine –K
Arginine – R
Phenylalanine – F

**68.** A. $C_6H_5COOH$ B. $C_6H_5COCl$ C. $C_6H_5COOH$ **D.** $C_6H_5CHO$

**69.** Kolbey's reaction Aspirin

**70.** $\lambda\alpha\frac{h}{mv}\Rightarrow\lambda\alpha\ h/p\Rightarrow\lambda=\frac{1}{p}$

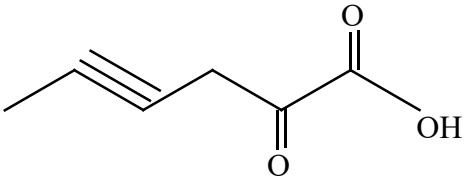
**71.** **3**

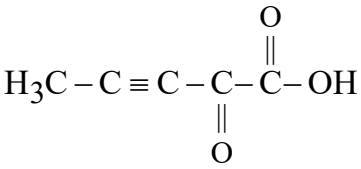


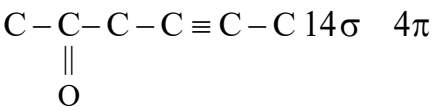

**72.** i→brown iv →Prussian blue

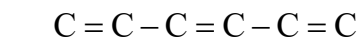
**73.** $CH_3CHO+H_2N.NH.CONH_2\rightarrow CH_3CH=N.NH.CO.NH_2$

**74.** **18+18 = 36**

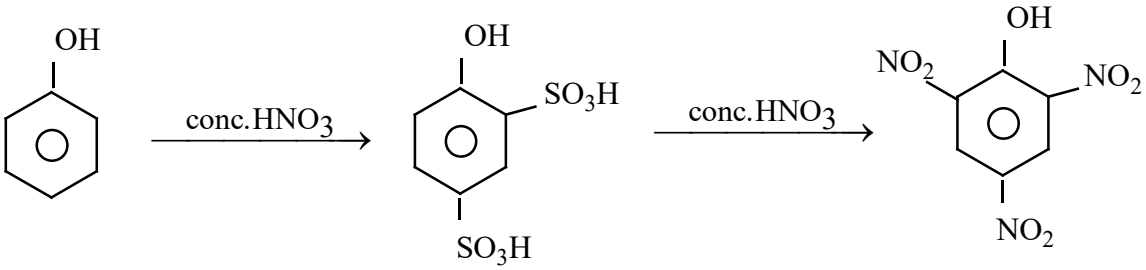



**75.**




(A) (B)